1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

After opening the cabinet the following components are found:

(a) **Motherboard:** Main circuit board connecting all the components and allows them to communicate and function together
(b) **CPU:** It is known as the brain of the computer. It takes instructions and executes it by performing calculations for all task.
(c) **RAM:** It is volatile memory that provides temporary storage for data and active programs which ensures all the multiple active tasks will run smoothly.
(d) **Storage (HDD/SDD):** It provides long term storage for all kind of system or application software along with personal files.
(e) **GPU:** Handles graphical performance for gaming and editing
(f) **Power Supply Unit:** Converts electrical power into safe, usable voltages required by the internal PC components.
(g) **Cooling System:** Releases heat generated by the hardware ensuring smooth running of all the components by maintaining temperature.

2. Take clear photos or draw labelled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step
Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Step-1: Switch off the power supply and unplug all external cable including power, keyboard, mouse and ethernet cable from the back of the case.
Step-2: open the case by opening the screws holding the panel and gently pulled the panel off.
Step-3: unplug all internal wires from the power supply that includes the 24 pin power connector , CPU power cable, SATA cable and smaller front panel headers.
Step-4: Gently press down on the plastic retaining clip on both sides of the RAM slot on the motherboard. The memory module pops up slightly, allowing me to lift it out.
Step-5: Disconnect the cables connecting with HDD and SSD. Unscrew it and take it out from the case.
Step-6: Remove the screws holding the motherboard inside the metal tray then tilt and gently lifts it out of the case.

→Due to the unavailability of the desktop PC to disassemble following given link can be used as the reference.
https://www.instructables.com/Disassemble-a-Computer/

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

https://www.youtube.com/watch?v=r5Esw32VkDI

Step-1: **Install the CPU:** Align the golden triangle on the processor with the corresponding corner mark on the socket. Lock the retention arm in place.
→This is the most critical and delicate component. Doing it outside the case prevents accidental pressure or drops onto delicate motherboard pins

Step-2: Remove the pre-installed heat spreader, align the SSD's gold connectors with the M.2 slot at a 30° angle, push it flat, and secure it with the screw.
→ Large CPU coolers or graphics cards will cover these slots once the board is in the case, making installation difficult later.

Step-3: Open the latches on the memory slots and firmly press the RAM sticks down until they click into place.
→ RAM can require significant downward force; installing it while the board is inside the case can flex the motherboard too much.

Step-4: Apply a small pea-sized drop of thermal paste (unless pre-applied) and mount the cooler or water block over the CPU.
→ Fitting a large heatsink inside a cramped PC case is much harder than doing it on an open table

Step-5: Pop the rear I/O shield into the back of the case and ensure the motherboard standoffs align with your specific board layout.

Step-6: Carefully lower the prepared motherboard into the case and secure it with the corresponding screws.
→ This needs to happen before bulky power supplies or graphics cards restrict your physical access to the case's edges

Step-7: Slide the PSU into the case's designated bay (usually the bottom) and secure it with screws.

Step-8: Plug the 24-pin power cable, CPU power cable, front panel headers (power button, reset, audio), and storage drives into the motherboard.
→ Connecting these tiny, fiddly headers becomes infinitely harder once you install a massive graphics card that blocks the lower half of the motherboard.

Step-9: Remove the corresponding PCIe slot covers from the back of the case. Push the GPU into the top PCIe slot until it clicks, and screw its bracket to the case.
→ Because of its sheer size and weight, the GPU will block your access to everything else on the motherboard. It should always be the last component installed inside the case

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

(a) Screwdriver for installing and removing screws without dropping them into tight, hard-to-reach areas
(b) Thermal Paste is required between the CPU and the heat sink for proper heat transfer and cooling.
(c) Zip Ties or Velcro Straps helps in manage cables, and preventswires from getting caught in fans.
(d) Small Bags/Containers are important for storing specific screws (like motherboard, case, and SSD screws) so they don't get lost